

# Covering 1024 syndromes with 50 columns

Grok Bot (4.6), Opus 5, GPT Sol 5.6

with instructions from Stephen Wu

16 August 2026

## Abstract

A binary linear  $[50, 40]_2$  code of covering radius 2 certifies  $\ell_2(10, 2) \leq 50$ , one column below the Kaikkonen–Rosendahl length 51 that has stood since 2003 and that still seeds the  $R = 2$  family of Davydov–Marcugini–Pambianco (arXiv:2511.02542). The new matrix admits a  $(2, 0)$ -partition with  $p(\mathbf{H}) = 10$ , so Construction  $\text{QM}_2^2$  propagates it to exhaustively verified codes of lengths 815 and 1631 at  $r = 18$  and  $r = 20$ , and to the family  $n = 51 \cdot 2^{r/2-5} - 1$  of asymptotic density 2601/2048. This note explains the problem, how the matrix was found, what was proved, and how to re-check every number from the committed text.

## Contents

|          |                                                           |          |
|----------|-----------------------------------------------------------|----------|
| <b>1</b> | <b>The problem</b>                                        | <b>2</b> |
| 1.1      | Covering radius . . . . .                                 | 2        |
| 1.2      | Why the $r = 10$ cell is load-bearing . . . . .           | 2        |
| <b>2</b> | <b>The process</b>                                        | <b>2</b> |
| 2.1      | Overnight search . . . . .                                | 2        |
| 2.2      | What the overnight run missed . . . . .                   | 3        |
| <b>3</b> | <b>The result</b>                                         | <b>3</b> |
| 3.1      | The code . . . . .                                        | 3        |
| 3.2      | Minimality . . . . .                                      | 3        |
| 3.3      | A $(2, 0)$ -partition with $p(\mathbf{H}) = 10$ . . . . . | 4        |
| 3.4      | The dependent triple . . . . .                            | 4        |
| 3.5      | Propagation . . . . .                                     | 4        |
| 3.6      | The family . . . . .                                      | 4        |
| <b>4</b> | <b>Verification</b>                                       | <b>5</b> |
| 4.1      | How to re-run . . . . .                                   | 5        |
| 4.2      | Two verifiers, opposite algorithms . . . . .              | 5        |
| 4.3      | Encoding . . . . .                                        | 5        |
| 4.4      | Independence, honestly . . . . .                          | 5        |
| <b>5</b> | <b>What is not claimed</b>                                | <b>6</b> |
| <b>6</b> | <b>Where the files live</b>                               | <b>6</b> |
| <b>A</b> | <b>The matrix as bits</b>                                 | <b>6</b> |

# 1 The problem

## 1.1 Covering radius

Let  $C \leq \mathbb{F}_2^n$  be a binary linear code of codimension  $r$ , so  $|C| = 2^{n-r}$ . Its *covering radius* is the least  $R$  such that every vector of  $\mathbb{F}_2^n$  lies at Hamming distance at most  $R$  from some codeword. Equivalently, if  $\mathbf{H}$  is an  $r \times n$  parity-check matrix with column set  $S \subset \mathbb{F}_2^r$ , every syndrome is a sum of at most  $R$  columns of  $\mathbf{H}$ . Write  $\ell_2(r, R)$  for the least such length.

A binary linear code of length  $n$  and dimension  $k = n - r$  is written  $[n, k]_2$ ; when the covering radius is  $R$  we append it,  $[n, k]_2 R$ . So a  $[50, 40]_2 2$  code is binary of length 50 and dimension 40, has covering radius 2, and hence redundancy  $r = 10$ .

For  $R = 2$  the condition is

$$\{0\} \cup S \cup (S + S) = \mathbb{F}_2^r. \quad (1)$$

A  $(2, 0)$ -partition of the columns (Definition 3.2 of [1]) is a partition into nonempty blocks such that every syndrome, including 0, is a sum of at most two columns from *distinct* blocks. Write  $p(\mathbf{H})$  for the number of blocks.

The finite covering density of an  $[n, n - r]_2 2$  code is the exact rational

$$\mu = \frac{1 + n + \binom{n}{2}}{2^r}.$$

Green’s Problem 40 asks for the asymptotic density

$$f(2) = \liminf_{r \rightarrow \infty} \mu(\ell_2(r, 2), r).$$

For an infinite family of  $[n(r), n(r) - r]_2 2$  codes, write  $\bar{\mu}(2)$  for the  $\liminf$  of  $\mu$  along the family as  $r \rightarrow \infty$ . Green’s  $f(2)$  takes the  $\liminf$  over the *optimal* lengths  $\ell_2(r, 2)$ , so any family bound  $\bar{\mu}(2) \leq c$  gives  $f(2) \leq c$ . It is not even known whether  $f(2) = 1$ . The pre-2025 upper bound  $f(2) \leq 729/512 \approx 1.42383$  had stood since 1992. Davydov, Marcugini and Pambianco [1] brought it to  $2704/2048 \approx 1.32031$  in November 2025, by iterating a construction they call  $\text{QM}_2^2$  from a single seed.  $\text{QM}_2^2$  grows a short radius-2 covering code into a longer one, provided the seed admits a  $(2, 0)$ -partition into few enough blocks; its exact input–output formula is recalled in Section 3.5.

## 1.2 Why the $r = 10$ cell is load-bearing

Their Table 5.1, row  $r = 10$ , still lists  $n = 51$ : the Kaikkonen–Rosendahl [2] code of 2003, density  $1327/1024 \approx 1.29590$ . The entry is not bolded. Even the 2025 paper did not improve it.

Their infinite family (Theorem 5.7) has lengths

$$\Phi(r) = 26 \cdot 2^{r/2-4} - 1 = 2^m(51 + 1) - 1, \quad m = r/2 - 5.$$

The 52 is the Kaikkonen–Rosendahl 51, plus one. The whole family is that one table cell pushed forward. Theorem 5.7(i) says the 2-partitions of  $\mathbf{H}_{KR}$  are “very important for the next iterative process”.

So a 50-column matrix at  $r = 10$  is not a one-cell footnote. If it admits a small enough  $(2, 0)$ -partition, it re-seeds the family. The paper’s computer search found  $p(\mathbf{H}_{KR}) = 11$ .

# 2 The process

## 2.1 Overnight search

On 16 August 2026 an automated agent (Codex `gpt-5.6-so1`, Max) ran a targeted simulated anneal seeded from the Kaikkonen–Rosendahl 51-column matrix. The state was a set of exactly

50 nonzero ten-bit columns. An array stored, for every syndrome, how many times it appeared as 0, a singleton, or a pair-sum. Swapping one column changes only its singleton and the 49 pair-sums involving it.

Most proposals were aimed at an uncovered syndrome  $g$ : offer  $g$  itself, or  $g + s$  for a current column  $s$ . With a fixed xorshift64 seed, run 0 reached zero uncovered syndromes at proposal 3,600,281.

The same search failed to close prettier one-column gaps:  $(r, n) = (8, 25)$  stalled at 3 uncovered syndromes of 256;  $(9, 38)$  stalled at 8 of 512;  $n = 49$  at  $r = 10$  stalled at 7 of 1024. Those are search residues, not lower bounds.

## 2.2 What the overnight run missed

The overnight folder certified  $\ell_2(10, 2) \leq 50$  and stopped at the table cell. It never computed a 2-partition, never ran  $\text{QM}_2^2$ , and still quoted the stale  $f(2) \leq 1.4238$ .

A later session (Claude, afternoon of the same day) treated the matrix as a claim to be verified, not as a trusted input: reconstructed the Kaikkonen–Rosendahl baseline from the hex in Theorem 4.3 of [1], found a  $(2, 0)$ -partition with  $p(\mathbf{H}) = 10$ , implemented  $\text{QM}_2^2$  from equations (4.2) and (4.4), and exhaustively checked the  $r = 18$  and  $r = 20$  outputs. Two verifiers were written from scratch in two languages, then a third quarantined oracle was run once as a post-hoc diff. The search procedure is not part of the argument. The argument is a finite decidable statement about committed matrices.

## 3 The result

### 3.1 The code

| claim            | value                                                                 |
|------------------|-----------------------------------------------------------------------|
| shape            | $10 \times 50$ , $\mathbb{F}_2$ -rank 10, 50 distinct nonzero columns |
| coverage         | 1024/1024 syndromes                                                   |
| covering radius  | exactly 2                                                             |
| density $\mu$    | $319/256 = 1.24609375$                                                |
| minimum distance | $d = 3$                                                               |
| KR baseline      | 51 columns, 1024/1024, density 1327/1024                              |

The columns, as unsigned integers with bit  $i$  (LSB first) equal to row  $i + 1$ :

|     |     |     |     |     |     |     |     |      |      |
|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| 1   | 2   | 4   | 15  | 16  | 32  | 65  | 86  | 128  | 173  |
| 183 | 202 | 212 | 247 | 256 | 297 | 320 | 329 | 341  | 366  |
| 373 | 381 | 391 | 403 | 438 | 460 | 479 | 491 | 502  | 559  |
| 576 | 608 | 653 | 734 | 742 | 754 | 771 | 777 | 789  | 821  |
| 846 | 855 | 869 | 881 | 893 | 897 | 927 | 981 | 1003 | 1004 |

The multiplicity histogram of the  $1276 = 1 + 50 + \binom{50}{2}$  representations over the 1024 syndromes is  $\{1:859, 2:129, 4:24, 5:9, 6:3\}$ . There are no zeroes. Exactly 973 syndromes are neither 0 nor a column, so the radius is not 1.

### 3.2 Minimality

Deleting any one of the 50 columns leaves at least 9 syndromes uncovered. The best deletions — columns 381, 479 and 927 — leave exactly 9. So the 50-set is a minimal 1-saturating set in  $PG(9, 2)$ , equivalently a locally optimal (LO) covering code (the geometric name for a covering-radius-2 column set with no redundant column). It is not padded.

### 3.3 A $(2, 0)$ -partition with $p(\mathbf{H}) = 10$

| block | columns                                    |
|-------|--------------------------------------------|
| 0     | 2, 128, 202, 212, 771, 855, 897, 981       |
| 1     | 86                                         |
| 2     | 381, 893, 1003                             |
| 3     | 183, 297                                   |
| 4     | 1, 65, 247, 256, 320, 438, 502, 734        |
| 5     | 15, 173, 329, 366, 460, 559, 653, 846      |
| 6     | 4, 16, 391, 491, 742, 754, 869, 881        |
| 7     | 479, 1004                                  |
| 8     | 403                                        |
| 9     | 32, 341, 373, 576, 608, 777, 789, 821, 927 |

All 973 pair-needed syndromes have a cross-block pair. The pair-only histogram over those 973 is  $\{1:821, 2:123, 4:19, 5:8, 6:2\}$ , so 821 syndromes have a unique pair and force their two columns into distinct blocks. Against the paper's  $p(\mathbf{H}_{KR}) = 11$ .

### 3.4 The dependent triple

There are exactly 10 linearly dependent triples of columns. Exactly one,  $(491, 734, 821)$ , lies in three distinct blocks (blocks 6, 4, 9). This is the analogue of [1] Theorem 5.2(ii), the ingredient Construction  $\text{QM}_5^2$  needs at  $r = 28$ . That construction is *not* carried out here.

### 3.5 Propagation

Construction  $\text{QM}_2^2$  ([1], Theorem 4.1) takes an  $[n_0, n_0 - r_0]_2$  code with a 2-partition into  $p(\mathbf{H}_0)$  blocks, an indicator set  $\mathcal{B} = \mathbb{F}_{2^m}$  subject to  $n_0 \geq 2^m \geq p(\mathbf{H}_0)$ , and  $\mathbf{D} = \mathbf{D}_1(2)$ , and returns

$$n = 2^m(n_0 + 1) - 1, \quad r = r_0 + 2m, \quad p(\mathbf{H}_C) \leq 2^{m+1} + 1.$$

With  $n_0 = 50$  and  $p(\mathbf{H}_0) = 10$  the inequality permits exactly  $m = 4$  and  $m = 5$  ( $2^6 = 64 > 50$ ). Both outputs were enumerated completely:

|         | $r$ | $n$  | coverage        | density         | published    |
|---------|-----|------|-----------------|-----------------|--------------|
| seed    | 10  | 50   | 1024/1024       | 319/256         | 51 (KR 2003) |
| $m = 4$ | 18  | 815  | 262144/262144   | 332521/262144   | 831          |
| $m = 5$ | 20  | 1631 | 1048576/1048576 | 1330897/1048576 | 1663         |

A second pair of matrices, from a different indicator allocation, also verifies (see `result/data/alt/`). Indicator assignment is a free choice, so two allocators give two codes.

### 3.6 The family

Iterating from  $(r, p) = (10, 10)$  with the paper's bound  $p(\mathbf{H}_C) \leq 2^{m+1} + 1$ , the even  $r \leq 64$  reachable by  $\text{QM}_2^2$  are

$$10, 18, 20, 30, 32, 34, 36, 38, 40, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64.$$

The even  $r$  *not* reachable this way are 12, 14, 16, 22, 24, 26, 28, 42, 44. Those gaps are real. The paper fills its own  $r = 22, 24, 26$  with  $\text{QM}_3^2$  and  $r = 28$  with  $\text{QM}_5^2$ ; the analogues are not implemented here.

Closed form on the reachable even  $r$ :

$$n = 51 \cdot 2^{r/2-5} - 1, \quad \mu = \frac{51^2}{2^{11}} - \frac{51}{2^{r/2+1}} + \frac{1}{2^r} \nearrow \frac{2601}{2048} \approx 1.27002.$$

Hence  $\bar{\mu}(2) \leq 2601/2048$ , and since  $f(2)$  is a lim inf over  $r$ , also  $f(2) \leq 2601/2048$ . Against the literature:

|                  | $\bar{\mu}(2)$              | source                  |
|------------------|-----------------------------|-------------------------|
| pre-2025         | $729/512 \approx 1.42383$   | standing since 1992     |
| arXiv:2511.02542 | $2704/2048 \approx 1.32031$ | seeded by KR $n_0 = 51$ |
| this seed        | $2601/2048 \approx 1.27002$ | seeded by $n_0 = 50$    |

Rows past  $m = 5$  inherit the paper’s  $p(\mathbf{H}_C)$  bound rather than computed partitions. That is the weakest link.

## 4 Verification

### 4.1 How to re-run

```
cd problems/covering/result
./run_all.sh
```

Needs `python3` and `rustc`. No network, no packages, no RNG. Seventy-four named assertions, exit 0, output byte-identical across runs (about two seconds).

### 4.2 Two verifiers, opposite algorithms

`verify/verify.py` is pair-driven: walk all  $\binom{n}{2}$  pairs, mark the syndromes they hit, then sweep  $2^r$  for anything unmarked. `verify/verify.rs` is syndrome-driven: for each of the  $2^r$  syndromes, scan the columns and test membership of  $s \oplus h$ . It then runs its own pair loop and refuses to report unless the two verdicts agree. They also differ in parser, rank pivot (lowest vs highest set bit), and exact rational arithmetic. `run_all.sh` diffs their fact dumps.

Both read the matrix from *text*. Neither consults a stored certificate. The partition JSON is used only for block labels; the columns themselves are re-derived from the matrix.

### 4.3 Encoding

Everything in the repo is LSB-first: bit  $i$  of a column integer is row  $i + 1$ . The hex listing of  $M_{KR}$  in [1] Theorem 4.3 is the opposite (MSB-first, row 1 as the high bit). Reconstructing  $\mathbf{H}_{KR} = [I_{10} \mid M_{KR}]$  reverses the ten bits. A from-scratch verifier that misses the reversal fails on the Kaikkonen–Rosendahl baseline *and only there*, which is a useful signal. The reversal is pinned by the paper’s own Theorem 5.2(ii),  $h_5 + h_{27} + h_{29} = 0$ , asserted in the builder as a canary.

A reader using the opposite convention gets the bit-reverse of every column, an  $\mathbb{F}_2$ -linear relabelling of  $\mathbb{F}_2^{10}$ , and every claim here is preserved.

### 4.4 Independence, honestly

Both verifiers were written in one session by one author. Two languages, two algorithms, two parsers, no shared code — but not two people. A third, quarantined oracle was run once after both were green and agreed everywhere; it is not in the tree.

## 5 What is not claimed

- 50 is not shown optimal. Sphere-covering gives only  $\ell_2(10, 2) \geq 45$ , since  $1+n+\binom{n}{2} \geq 1024$  fails at  $n = 44$ , which gives only 991. Minimality says this particular 50-set has no redundant column, which says nothing about a different 49-set.
- The  $n = 49$  run leaving 7 uncovered syndromes is search residue, not a lower bound.
- The gaps (8, 25) and (9, 38) remain open.
- Only  $m = 4$  and  $m = 5$  are exhaustively verified. The rest of the family inherits  $p(\mathbf{H}_C) \leq 2^{m+1} + 1$  from [1] (4.4).
- Nothing is claimed for  $r \in \{12, 14, 16, 22, 24, 26, 28, 42, 44\}$ .
- $f(2) \leq 2601/2048$  is an upper bound only. Whether  $f(2) = 1$  is untouched.
- Priority is unresolved. Table 5.1 of [1] is “best as far as the authors know”. A  $\ell_2(10, 2) \leq 50$  may already exist in ACCT proceedings. See `result/PRIORITY.md`.

## 6 Where the files live

This document is self-contained. The paths below matter only to a reader who also has the repository.

| path                                        | what                                        |
|---------------------------------------------|---------------------------------------------|
| <code>result/NOTE.md</code>                 | repo companion, for a referee with the tree |
| <code>result/note.tex</code>                | arXiv-style draft of the same claims        |
| <code>result/run_all.sh</code>              | the pipeline                                |
| <code>result/data/H_r10_n50.txt</code>      | the matrix                                  |
| <code>result/data/partition_p10.json</code> | the 10-block partition                      |
| <code>explainer.html</code>                 | interactive companion                       |
| <code>WALKTHROUGH.md</code>                 | how the overnight search felt               |
| <code>ATTACK.md</code>                      | chronological log                           |

## A The matrix as bits

Rows 1 to 10 from top to bottom, columns left to right. Bit  $i$  (LSB) of each integer column is row  $i + 1$ .

```

10010010011001010110111100110100100011110111111110
01010001001101000001001110111100011110001100001010
00110001011011000011111011101100111000111110101101
00010000010100010101010001110100110001001000101011
00001001001011000010110110101000010100110101101100
00000100011001010001110010011101001100010011100011
00000011000111001111110001111011011100001111100111
0000000011111100000000111111000111100000000011111
000000000000000111111111111100000001111111111111
000000000000000000000000000000011111111111111111

```

## References

- [1] A. A. Davydov, S. Marcugini, F. Pambianco, *New upper bounds for binary linear covering codes*, arXiv:2511.02542 [cs.IT], November 2025.
- [2] M. Kaikkonen, C. Rosendahl, *New covering codes from an ADS-like construction*, IEEE Trans. Inform. Theory **49** (2003), no. 7, 1809–1812.
- [3] G. Cohen, I. Honkala, S. Litsyn, A. Lobstein, *Covering Codes*, North-Holland Mathematical Library **54**, Elsevier, Amsterdam, 1997.
- [4] B. Green, *100 open problems*, Problem 40. <https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf>.